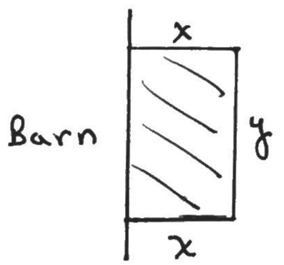
Perimeter, Surface Area, Volume Questions

Example 1: A farmer wants to build a small pen for his chickens. One side of the pen will be against his barn. If he has 80 metres of fencing to use, what is the maximum possible area?

Solution:

Let x be the width of the pen.
Let y be the length of the pen.



$$P = 2x + y$$
$$80 = 2x + y$$
$$y = 80 - 2x$$

$$P(x) = xy$$
$$= x(80 - 2x)$$
$$P'(x) = 80 - 4x$$
$$P'(x) = 0$$
$$80 - 4x = 0$$
$$\therefore x = 20 \Rightarrow y = 40$$

* check: $P''(20) = -4 < 0 \Rightarrow$ max.

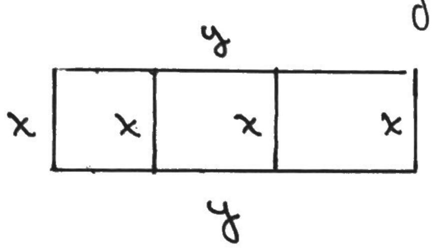
$\therefore$ Maximum area is $20\text{m} \times 40\text{m} = 800\text{m}^2$.

Example 2: A rancher wants to enclose 3 equal adjacent rectangular lots with an area of $600\ m^2$ each. What is the least amount of fencing required?

Solution:

Let x be the width of each lot.

Let y be the total length of the lots.



$$A = x \cdot y$$

$$1800 = x \cdot y$$

$$y = \frac{1800}{x}$$

$$P(x) = 4x + 2y$$

$$= 4x + \frac{3600}{x}$$

$$P'(x) = 4 - \frac{3600}{x^2}$$

$$P'(x) = 0$$

$$4 - \frac{3600}{x^2} = 0$$

$$x^2 = 900$$

$x = \pm 30$ (-30 is inadmissible)

$$y = 60$$

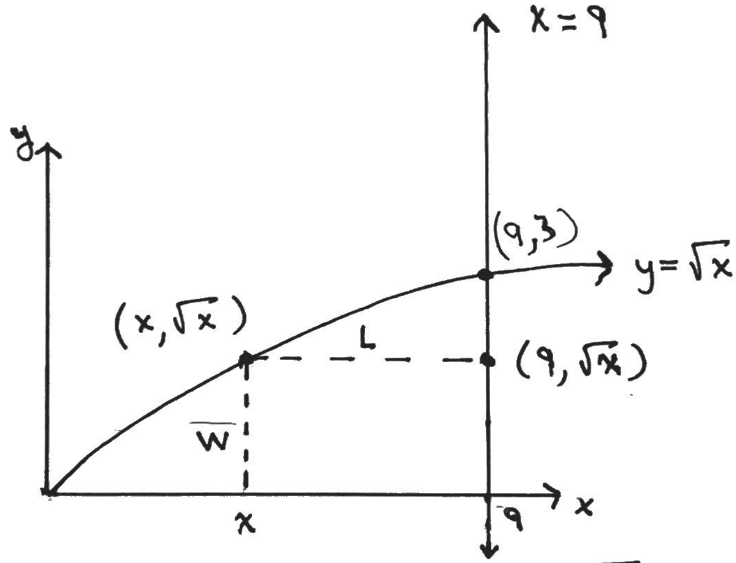
* check: $P''(30) = \frac{7200}{(30)^3}$

$\doteq 0.27 > 0 \Rightarrow$ min.

$\therefore$ The amount of fencing is $4(30) + 2(60) = 240\,m$

Example 3: Determine the dimensions of the rectangle of largest area which can be inscribed in the closed region bounded by the x-axis, $x = 9$ and $y = \sqrt{x}$.

Solution:



$\text{Length} = 9 - x$; $\text{width} = \sqrt{x} - 0$
$= \sqrt{x}$

$A = L \cdot W$

$A = (9-x)\sqrt{x}$

$A' = -\sqrt{x} + \dfrac{9-x}{2\sqrt{x}}$

$A' = 0$

$-\sqrt{x} + \dfrac{9-x}{2\sqrt{x}} = 0$

$-2x + 9 - x = 0$

$3x = 9$

$x = 3$

$\therefore\ W = \sqrt{x}$; $L = 9 - x$

$W = \sqrt{3}$ $\qquad L = 9 - 3$

$L = 6$

$\therefore\ \text{Area} = 6 \times \sqrt{3}\ \text{units}^2$.

Example 4: A manufacturer wishes to produce cylindrical fruit cans with a capacity of 250 mL. What dimensions will minimize the amount of material required for the can?

Solution:

$V = 250 \text{ mL}$

$250 = \pi r^2 h$

$h = \dfrac{250}{\pi r^2}$

$$S.A. = 2\pi r^2 + 2\pi r h$$

$$S.A. = 2\pi r^2 + 2\pi r\left(\frac{250}{\pi r^2}\right)$$

$$S.A. = 2\pi r^2 + \frac{500}{r}$$

$$S.A.' = 4\pi r - \frac{500}{r^2}$$

$$S.A.' = 0$$

$$4\pi r - \frac{500}{r^2} = 0$$

$$4\pi r^3 - 500 = 0$$

$$r^3 = \frac{500}{4\pi}$$

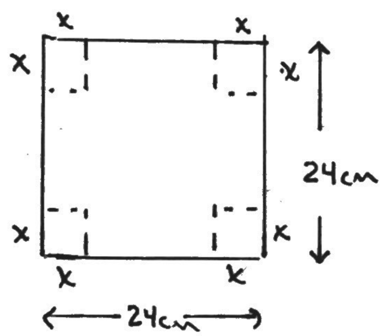
$$r = \sqrt[3]{\frac{500}{4\pi}} \doteq 3.41 \text{ cm}$$

$$\Rightarrow h \doteq 6.83 \text{ cm}$$

$\therefore$ $r \doteq 3.41$ cm and $h \doteq 6.83$ cm.

Example 5: An open topped box is to be made from a square piece of cardboard that is 24 cm x 24 cm by cutting equal squares from each corner and folding up the flaps to make sides. What is the greatest volume for the box?

Solution:



Let x = height

Let $24-2x$ = length = width

$V = L \cdot w \cdot h$

$V = (24-2x)^2 \cdot x$

$V' = 2(24-2x)(-2)(x) + (24-2x)^2$

$V' = (24-2x)[-4x + (24-2x)]$

$V' = (24-2x)(-6x+24)$

$V' = -12(12-x)(x-4)$

$V' = 0$

$-12(12-x)(x-4) = 0$

$\therefore$ $x = 12$, $x = 4$

inadmissible.

$\therefore$ $h = 4$ cm

$L = W = 16$ cm

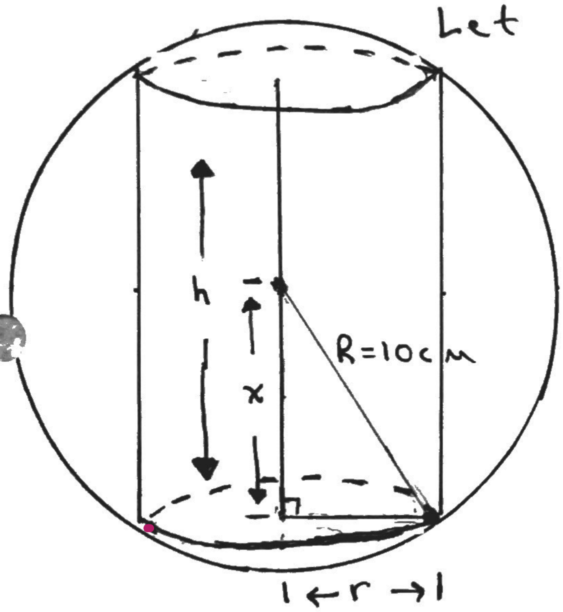
Volume $= 16_{cm} \times 16_{cm} \times 4_{cm}$

Volume $= 1{,}024$ cm^3

Example 6: Determine the dimensions of the cylinder of maximum volume that can be inscribed in a sphere of radius 10 cm.

Solution: Let R = radius of the sphere.

Let r = radius of the cylinder.



$$R^2 = x^2 + r^2$$

$$r = \sqrt{R^2 - x^2}$$

$$r = \sqrt{100 - x^2}$$

note: $h = 2x$

$$V = \pi r^2 h$$

$$V = \pi\left(\sqrt{100 - x^2}\right)^2 (2x)$$

$$V = \pi(100 - x^2)(2x)$$

$$V = \pi(200x - 2x^3)$$

$$V' = \pi(200 - 6x^2)$$

$$V' = 0$$

$$\pi(200 - 6x^2) = 0$$

$$x^2 = \frac{100}{3}$$

$$x = \pm\frac{10}{\sqrt{3}} \quad \left(-\frac{10}{\sqrt{3}} \text{ is inadmissable}\right)$$

$\therefore$ $h = \frac{20}{\sqrt{3}}$ cm and

$$r = \sqrt{100 - \frac{100}{3}} = \sqrt{\frac{200}{3}} \text{ cm.}$$